

AICE3001/AICE6002: Mathematics of Machine Learning

Subsidiary Notes — Lecture 26: Entropy

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1 Keywords

- Shannon entropy, bits and nats, code length, Huffman coding, Stirling's approximation, maximum entropy

2 Main Points

2.1 Measuring Uncertainty

- Which is more uncertain: three coin tosses or one throw of a die? It depends on whether we care about the order of the tosses—but how would we answer at all?
- For a discrete random variable X , Shannon's entropy gives a precise answer,

$$H_X = - \sum_{x \in \mathcal{X}} \mathbb{P}[X = x] \log_2(\mathbb{P}[X = x])$$

- **A fair die**, $\mathbb{P}[D = i] = 1/6$:

$$H_D = - \sum_{i=1}^6 \frac{1}{6} \log_2\left(\frac{1}{6}\right) = \log_2(6) \approx 2.585 \text{ bits}$$

- **Three ordered coin tosses**, eight equally likely outcomes: $H_C = \log_2(8) = 3$ bits. More equally likely outcomes means more uncertainty, as we'd expect
- **Three unordered coin tosses** (we only record the number of heads): the outcomes 0 and 3 heads have probability $1/8$ each, and 1 and 2 heads have probability $3/8$ each, so

$$H_U = -\frac{1}{4} \log_2\left(\frac{1}{8}\right) - \frac{3}{4} \log_2\left(\frac{3}{8}\right) \approx 1.811 \text{ bits}$$

- This seems reasonable, but it is not obvious how we would have arrived at a number without entropy. Why this particular formula?

2.2 Additivity

- For two independent random variables, $\mathbb{P}[X, Y] = \mathbb{P}[X] \mathbb{P}[Y]$, so

$$\begin{aligned} H_{(X,Y)} &= - \sum_{x,y} \mathbb{P}[X = x] \mathbb{P}[Y = y] \log_2(\mathbb{P}[X = x] \mathbb{P}[Y = y]) \\ &= - \sum_{x,y} \mathbb{P}[X = x] \mathbb{P}[Y = y] (\log_2(\mathbb{P}[X = x]) + \log_2(\mathbb{P}[Y = y])) \\ &= - \sum_x \mathbb{P}[X = x] \log_2(\mathbb{P}[X = x]) - \sum_y \mathbb{P}[Y = y] \log_2(\mathbb{P}[Y = y]) = H_X + H_Y \end{aligned}$$

using $\sum_y \mathbb{P}[Y = y] = \sum_x \mathbb{P}[X = x] = 1$

- The uncertainty of two independent events adds. Shannon's entropy is one of very few functions with this property

2.3 Entropy as Code Length

- To communicate one of 2^n equally likely outcomes (say n coin tosses) we need a binary string of exactly n bits
- With 5 equally likely outcomes, 3 bits would do but would waste 3/8 of the possible codes. Batching 3 independent messages gives $5^3 = 125$ outcomes, which fit in 7 bits ($2^7 = 128$), wasting only 3/128: that is $7/3 \approx 2.33$ bits per message
- Batching more and more messages, the cost per message approaches $\log_2(5) \approx 2.32$ bits. In general N equally likely outcomes need $\log_2(N) = -\log_2(1/N)$ bits each
- When outcomes have different probabilities we can still find codes that use very close to $-\log_2(\mathbb{P}[X = x])$ bits for outcome x , by transmitting many messages. This code length measures the *surprise* of the outcome: rare outcomes cost more
- Shannon's entropy is the *expected code length*,

$$H_X = \mathbb{E}_X[-\log_2(\mathbb{P}[X])]$$

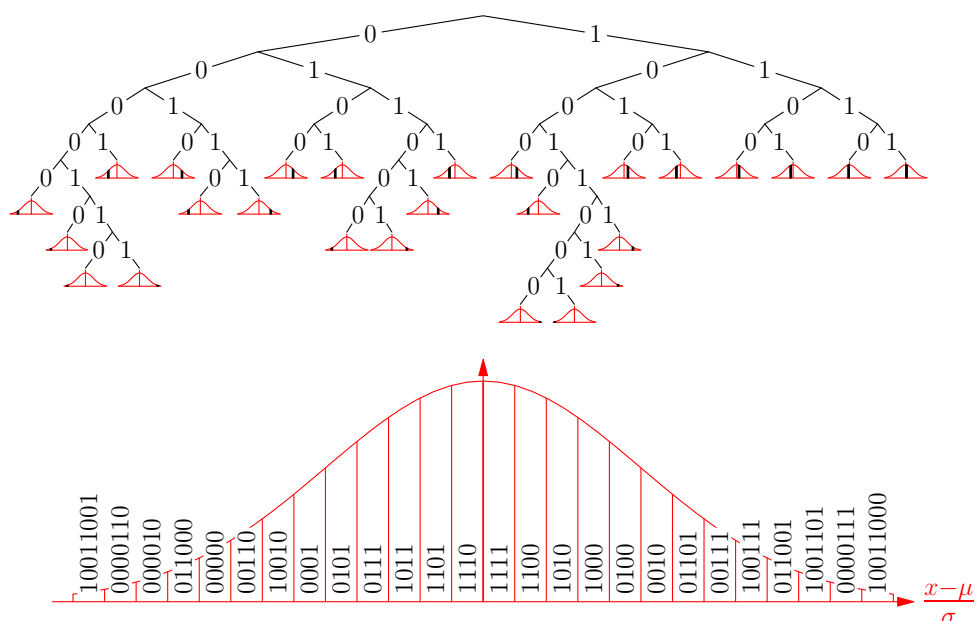
which is the average amount of information needed to convey the outcome

- The lecture animation illustrates this with a loaded die: each outcome's bar is the length $-\log_2(\mathbb{P}[X = x])$ of its code word, and the entropy is the probability-weighted average bar length

X_i :	1	2	3	4	5	6
$p(X_i)$:	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{4}$
Code:	000	001	010	011	10	11
$L = -\log_2(p(X_i))$:	3	3	3	3	2	2

2.4 Real Codes

- Code words of different lengths cause a practical problem: the receiver needs to know where each one ends
- *Huffman coding* solves this by placing the code words at the leaves of a binary tree, so no code word is a prefix of another and any message can be decoded by walking the tree
- A greedy algorithm (repeatedly merge the two least probable symbols) builds the optimal tree. Its expected code length L satisfies $H_X \leq L < H_X + 1$



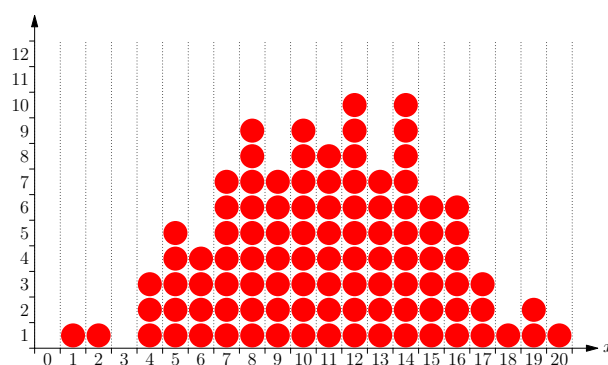
- The lecture illustrates coding a normal distribution discretised to accuracy Δx : finer accuracy means more possible values and a longer code

2.5 Bits and Nats

- With $\log_2$, entropy is measured in *bits*
- With natural logarithms, $H_X = -\sum_x \mathbb{P}[X = x] \ln(\mathbb{P}[X = x])$, entropy is measured in *nats*. One nat is $\log_2(e) \approx 1.443$ bits
- Nats are more convenient whenever we do calculus on entropy, which is why the rest of this note uses natural logarithms

2.6 Counting States

- Put N balls into K boxes so that n_i land in box i

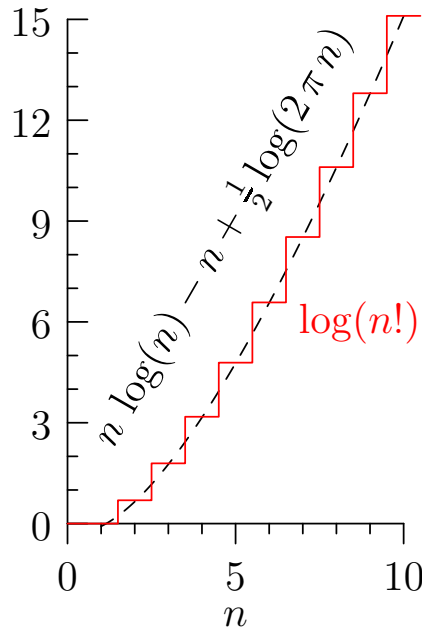


- The number of ways of doing this is the multinomial coefficient

$$W(\mathbf{n}) = \frac{N!}{n_1! n_2! \cdots n_K!}$$

- For large n , *Stirling's approximation* gives

$$n! \approx \sqrt{2\pi n} n^n e^{-n} \quad \log(n!) \approx n \log(n) - n + \frac{1}{2} \log(2\pi n)$$



- Substituting, the $-n$ terms cancel because $\sum_i n_i = N$, and to leading order

$$W(\mathbf{n}) = e^{\log(N!) - \sum_i \log(n_i!)} \approx C e^{N \log(N) - \sum_i n_i \log(n_i)} = C e^{-N \sum_i \frac{n_i}{N} \log(\frac{n_i}{N})}$$

- Writing $p_i = n_i/N$ for the fraction of balls in box i ,

$$W(\mathbf{n}) \approx C e^{N H} \quad \text{where} \quad H = - \sum_i p_i \log(p_i)$$

- So entropy measures the logarithm of the number of microscopic arrangements consistent with the proportions p_i . Because of the factor N in the exponent, as $N \rightarrow \infty$ the arrangements consistent with any constraints are overwhelmingly concentrated on the proportions that maximise the entropy

2.7 The Maximum Entropy Method

- To infer a distribution from limited information, maximise its entropy subject to constraints expressing what we know. The entropy acts as a prior
- Two justifications:
 - it picks by far the most likely set of configurations consistent with the observations
 - it maximises our uncertainty given what we know, making the estimate as unbiased as possible
- It relies on the underlying arrangements being equally likely a priori. If there are other constraints we have not included, it can give poor answers
- **Known mean and variance.** For a continuous X with $\mathbb{E}[X] = \mu$ and $\mathbb{E}[X^2] = \mu_2 = \mu^2 + \sigma^2$, form the Lagrangian

$$\begin{aligned} \mathcal{L}(f) = & - \int f_X(x) \log(f_X(x)) \, dx + \lambda_0 \left(\int f_X(x) \, dx - 1 \right) \\ & + \lambda_1 \left(\int f_X(x) x \, dx - \mu \right) + \lambda_2 \left(\int f_X(x) x^2 \, dx - \mu_2 \right) \end{aligned}$$

- Setting the functional derivative to zero,

$$\frac{\delta \mathcal{L}}{\delta f_X(x)} = -\log(f_X(x)) - 1 + \lambda_0 + \lambda_1 x + \lambda_2 x^2 = 0$$

so $f_X(x) = e^{-1+\lambda_0+\lambda_1 x+\lambda_2 x^2}$

- Choosing the multipliers to satisfy the three constraints gives

$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)}$$

The *normal distribution is the maximum entropy distribution with a given mean and variance*

- Maximum entropy works superbly in statistical physics, which infers distributions over microscopic states from observations such as volume and pressure. Temperature appears, rather strangely, as a Lagrange multiplier

2.8 A Little History

- Entropy was introduced in physics by Rudolf Clausius in 1865 (Macquorn Rankine discussed it in 1850)
- Ludwig Boltzmann interpreted it as counting the number of states; Josiah Willard Gibbs and James Clerk Maxwell completed the picture
- Claude Shannon founded information theory on entropy around 1948
- Ed Jaynes realised that statistical physics can be seen as an inference problem—the maximum entropy method

3 Exercises

3.1 The uniform distribution

- Using a Lagrange multiplier for normalisation, show that the maximum entropy distribution over K outcomes with no other constraints is uniform, and find its entropy

3.2 Knowing only the mean

- Find the maximum entropy density on $x \geq 0$ with a given mean $\mathbb{E}[X] = \mu$

3.3 Entropy of a biased coin

- Find the entropy $H(p)$ in bits of a coin that comes up heads with probability p . For which p is it largest? Evaluate $H(0.1)$

3.4 Finishing the normal derivation

- Show that the multipliers in the maximum entropy derivation must be $\lambda_2 = -1/(2\sigma^2)$ and $\lambda_1 = \mu/\sigma^2$

4 Answers

4.1 The uniform distribution

- Maximise $-\sum_i p_i \log(p_i) + \lambda (\sum_i p_i - 1)$. Differentiating with respect to p_i gives $-\log(p_i) - 1 + \lambda = 0$, so every $p_i = e^{\lambda-1}$ takes the same value
- Normalisation forces $p_i = 1/K$, with entropy $\log(K)$ nats or $\log_2(K)$ bits

4.2 Knowing only the mean

- Maximise $-\int_0^\infty f \log f \, dx + \lambda_0 (\int_0^\infty f \, dx - 1) + \lambda_1 (\int_0^\infty f x \, dx - \mu)$. The stationarity condition gives $f(x) = e^{-1+\lambda_0+\lambda_1 x}$
- Normalisability on $[0, \infty)$ needs $\lambda_1 < 0$; write $f(x) = A e^{-ax}$. Normalisation gives $A = a$, and the mean of $a e^{-ax}$ is $1/a$, so $a = 1/\mu$:

$$f(x) = \frac{1}{\mu} e^{-x/\mu}$$

the exponential distribution

4.3 Entropy of a biased coin

- $H(p) = -p \log_2(p) - (1-p) \log_2(1-p)$
- $H'(p) = \log_2((1-p)/p)$, which vanishes at $p = 1/2$, where $H = 1$ bit. $H''(p) < 0$, so this is the maximum
- $H(0.1) \approx 0.469$ bits

4.4 Finishing the normal derivation

- The log density is quadratic in x , $\log f_X(x) = -1 + \lambda_0 + \lambda_1 x + \lambda_2 x^2$
- The log of a normal density is $-(x - \mu)^2/(2\sigma^2) + \text{const} = -\frac{x^2}{2\sigma^2} + \frac{\mu x}{\sigma^2} + \text{const}$
- Matching coefficients of x^2 and x gives $\lambda_2 = -1/(2\sigma^2)$ and $\lambda_1 = \mu/\sigma^2$. The constant, and hence λ_0 , is then fixed by normalisation. This works because a normal with the right mean and variance satisfies all three constraints, and a quadratic exponent determines a normal distribution uniquely